Lecture 0

Mathematical preliminaries

Objectives. *An observable in infinite dimension may act only on a dense subspace. This lecture defines its domain, adjoint and self-adjointness. A diagonal example distinguishes symmetry from self-adjointness. Multiplication operators prepare the construction of position and momentum in Lecture 4.*

- Determine the adjoint domain of a diagonal operator.
- Distinguish symmetric operators from self-adjoint operators.
- Explain why a real continuous multiplier is self-adjoint on its maximal domain.

0.1 Operators that cannot act on every vector

This course builds the quantum theory of a particle. It starts in a box, goes on to the whole line, and ends with hydrogen. As in *Quantum Mechanics I*, it asks four questions about each system: what its states are, what its observables are, how it evolves, and what a measurement gives. From Lecture 1 on, a box near the end of each lecture collects the answers that the lecture gives. Figure 0.1 shows the lectures in their three Parts, some of the earlier lectures that each one uses, and the examples that return through the course. This lecture prepares the tools.

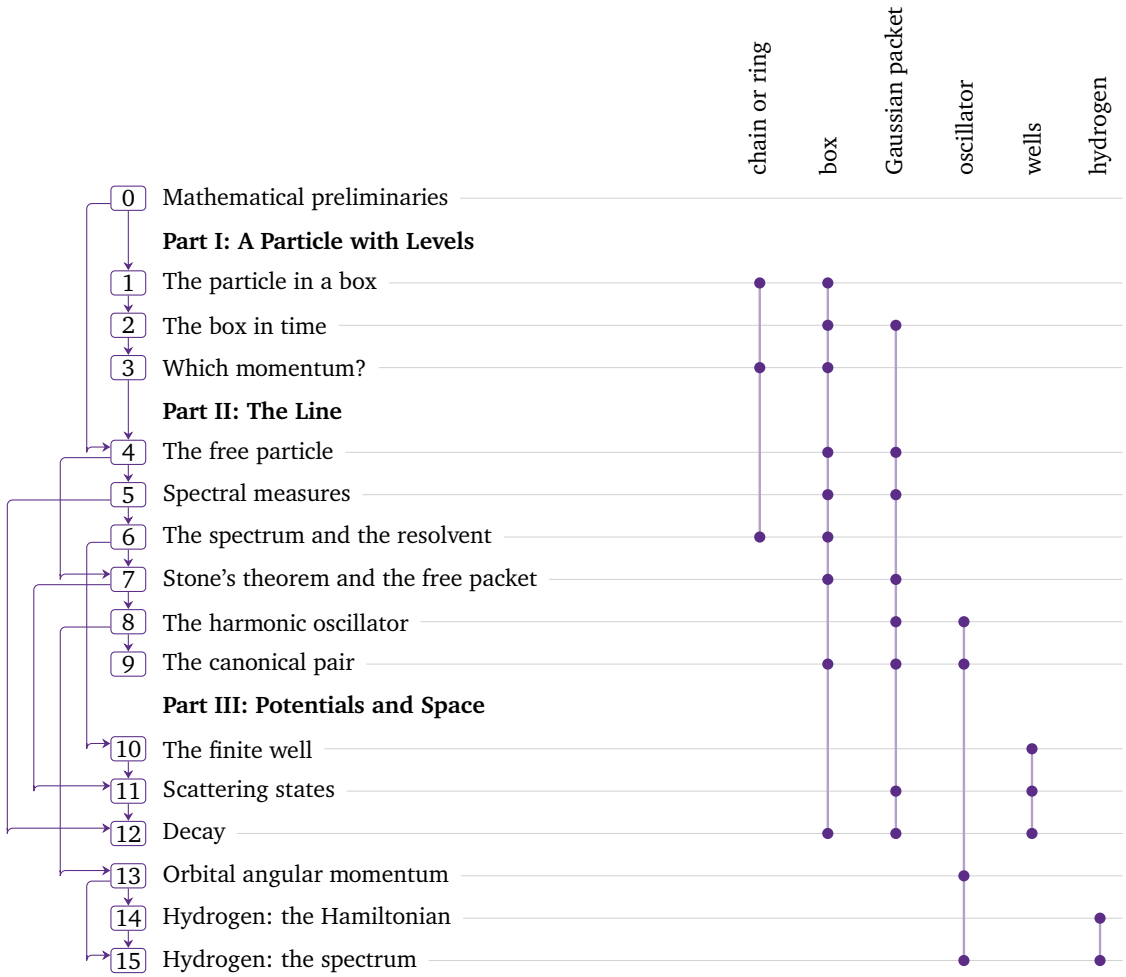


Figure 0.1: The lectures of the course in order, grouped by the three Parts. The titles are shortened. An arrow from one lecture to a later one means that the later lecture uses the earlier one. Only some of these arrows are drawn. A dot marks a lecture that works with the example named at the head of its column, and a line joins the dots of one example. The dots, like the arrows, are a selection.

The observables of a particle need domains. Its position Q and momentum P should satisfy the *canonical commutation relation* $QP\psi - PQ\psi = \mathrm{i}\hbar\psi$ for the vectors ψ on which both sides are defined. By *Quantum Mechanics I*, Prop. 9.8, no pair of $n \times n$ matrices does, since the trace of $QP - PQ$ is 0 and the trace of $\mathrm{i}\hbar\mathbb{1}$ is $\mathrm{i}\hbar n$. In infinite dimension two cited theorems take over. They concern *bounded* operators and the self-adjoint ones among them (*Linear Algebra*, Defs 14.2 and 14.3).

By Moretti, Prop. 11.32, no two bounded self-adjoint operators Q and P on a Hilbert space satisfy $QP - PQ = \mathrm{i}\hbar\mathbb{1}$.

By the *Hellinger–Toeplitz theorem* (Teschl, Thm 2.9), an operator A that is defined on every vector and symmetric, that is, $\langle\varphi, A\psi\rangle = \langle A\varphi, \psi\rangle$ for all φ and ψ, is bounded. Observables are symmetric, as in *Quantum Mechanics I*. So Q and P cannot both be defined on every vector, and they satisfy the canonical relation only on part of the space (Lecture 4).

The energy of a particle in a box is another example. Lecture 1 shows that it multiplies a sequence of unit vectors by numbers that grow without bound. So it is not bounded, and by the Hellinger–Toeplitz theorem it cannot be a symmetric operator defined on every vector. In infinite dimension, an operator comes with the set of vectors it acts on.

0.2 Operators with domains

Definition 0.1 (Operator; densely defined)**.** An *operator* A on a Hilbert space $\mathcal{H}$ is a linear map from a subspace $D(A) \subseteq \mathcal{H}$, its *domain*, into $\mathcal{H}$. It is *densely defined* if every vector of $\mathcal{H}$ is the limit of a sequence in $D(A)$. Two operators are equal when they have the same domain and the same action on it.

Definition 0.2 (Adjoint)**.** Let A be a densely defined operator on $\mathcal{H}$. Its *adjoint* $A^\dagger$ has the domain

$$D(A^\dagger) = \big\{\varphi \in \mathcal{H} : \text{ there is } \chi \in \mathcal{H} \text{ with } \langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle \text{ for every } \psi \in D(A)\big\}, \quad (0.1)$$

and on it $A^\dagger\varphi = \chi$.

The density of the domain makes χ unique. Suppose that χ_1 and χ_2 both satisfy the condition in (0.1), and let $v = \chi_1 - \chi_2$. Then $\langle v, \psi\rangle = 0$ for every $\psi \in D(A)$. Since $D(A)$ is dense, there is a sequence ψ_j in $D(A)$ with $\psi_j \to v$. By the Cauchy–Schwarz inequality,

$$\|v\|^2 = \big|\langle v, v\rangle - \langle v, \psi_j\rangle\big| = \big|\langle v, v - \psi_j\rangle\big| \le \|v\|\,\big\|v - \psi_j\big\| \longrightarrow 0.$$

So $v = 0$, and $\chi_1 = \chi_2$. This is where the definition needs A to be densely defined.

The adjoint is an operator. Indeed, let φ_1 and φ_2 lie in $D(A^\dagger)$, with the vectors χ_1 and χ_2, and let a and b be numbers. Since the inner product is antilinear in its first slot,

$$\langle a\varphi_1 + b\varphi_2, A\psi\rangle = \overline{a}\,\langle \chi_1, \psi\rangle + \overline{b}\,\langle \chi_2, \psi\rangle = \langle a\chi_1 + b\chi_2, \psi\rangle \quad \text{for every } \psi \in D(A).$$

So $a\varphi_1 + b\varphi_2$ lies in $D(A^\dagger)$, and $A^\dagger(a\varphi_1 + b\varphi_2) = a\chi_1 + b\chi_2$.

On $\mathbb{C}^n$ the adjoint is the conjugate transpose. More generally, if A is bounded and $D(A) = \mathcal{H}$, the adjoint of *Linear Algebra*, Def. 14.3, gives a vector χ for every φ, and $A^\dagger$ is that adjoint. For other operators χ need not exist for every φ, and $D(A^\dagger)$ records the φ for which it does.

Definition 0.3 (Symmetric; self-adjoint)**.** A densely defined operator A is *symmetric* if $\langle \varphi, A\psi\rangle = \langle A\varphi, \psi\rangle$ for all $\varphi, \psi \in D(A)$. It is *self-adjoint* if $A^\dagger = A$, that is, if $D(A^\dagger) = D(A)$ and $A^\dagger\varphi = A\varphi$ there.

If A is symmetric and $\varphi \in D(A)$, then $\chi = A\varphi$ satisfies the condition in (0.1). So $D(A) \subseteq D(A^\dagger)$, with $A^\dagger\varphi = A\varphi$ there. Self-adjointness asks in addition that the adjoint have no larger domain. On $\mathbb{C}^n$ the two notions coincide, and both are the self-adjoint matrices of *Quantum Mechanics I*.

The adjoint of a symmetric operator contains the operator. Containing is a relation between operators that the course uses often.

Definition 0.4 (Extension)**.** Let A and B be operators on $\mathcal{H}$. We write $A \subseteq B$, and say that B *extends* A, when

$$D(A) \subseteq D(B) \quad \text{and} \quad B\psi = A\psi \quad \text{for every } \psi \in D(A).$$

So a symmetric A satisfies $A \subseteq A^\dagger$, and it is self-adjoint when also $A^\dagger \subseteq A$.

Proposition 0.5 (Adjoints reverse inclusions)**.** *Let A and B be densely defined operators with $A \subseteq B$. Then $B^\dagger \subseteq A^\dagger$.*

Proof. Let $\varphi \in D(B^\dagger)$, with $\chi = B^\dagger\varphi$. Then $\langle \varphi, B\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(B)$. Every $\psi \in D(A)$ lies in $D(B)$, with $B\psi = A\psi$. So $\langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(A)$. Hence $\varphi \in D(A^\dagger)$, and $A^\dagger\varphi = \chi = B^\dagger\varphi$. ∎

A larger operator has to satisfy the condition (0.1) for more vectors ψ, so its adjoint has a smaller domain. The following example shows a symmetric operator whose adjoint is strictly larger.

Example 0.6 (A diagonal operator on finite sequences)**.** Work in the space ℓ^2 of square-summable sequences $x = (x_1, x_2, \dots)$, with $\langle x, y\rangle = \sum_n \overline{x_n}\, y_n$ (*Linear Algebra*, Def. 13.15). Let e_n be the sequence with 1 in place n and 0 elsewhere.

1. *The operator.* Let $D(D_0)$ be the finite sequences, those with only finitely many nonzero entries, and put
$$(D_0 x)_n = n\, x_n.$$
The truncations $(x_1, \dots, x_M, 0, 0, \dots)$ of any $x \in \ell^2$ are finite sequences, and they converge to x, since $\sum_{n>M} |x_n|^2 \to 0$. So D_0 is densely defined.
2. *Symmetric.* For finite sequences x and y, the sums are finite, and $\langle x, D_0 y\rangle = \sum_n \overline{x_n}\, n y_n = \langle D_0 x, y\rangle$.
3. *The adjoint.* Let $\varphi \in D(D_0^\dagger)$, with $\chi = D_0^\dagger\varphi$. Test the condition (0.1) on $\psi = e_n$. The left side is $\langle \varphi, D_0 e_n\rangle = n\overline{\varphi_n}$, and the right side is $\langle \chi, e_n\rangle = \overline{\chi_n}$. So $\chi_n = n\varphi_n$ for every n, and since $\chi \in \ell^2$,
$$\sum_n n^2\, |\varphi_n|^2 = \|\chi\|^2 < \infty.$$
Conversely, if $\sum_n n^2\, |\varphi_n|^2 < \infty$, then $\chi = (n\varphi_n)$ lies in ℓ^2 and $\langle \varphi, D_0\psi\rangle = \langle \chi, \psi\rangle$ for every finite ψ. Hence
$$D(D_0^\dagger) = \Big\{ \varphi \in \ell^2 : \sum_n n^2\, |\varphi_n|^2 < \infty \Big\}, \qquad (D_0^\dagger\varphi)_n = n\, \varphi_n.$$
4. *Not self-adjoint.* The sequence $\varphi_n = 1/n^2$ has $\sum_n n^2\, |\varphi_n|^2 = \sum_n 1/n^2 < \infty$, so it lies in $D(D_0^\dagger)$. It is not a finite sequence. So $D(D_0) \neq D(D_0^\dagger)$, and D_0 is symmetric and not self-adjoint.

The test on the vectors e_n in the third step fixed every coefficient of χ. Lecture 1 applies the same test to the energy of a particle in a box.

The operators D_0 and $D_0^\dagger$ act by the same rule, which multiplies the nth entry by n. They differ only in their domains. Figure 0.2 draws these domains inside ℓ^2 and places three sequences in them.

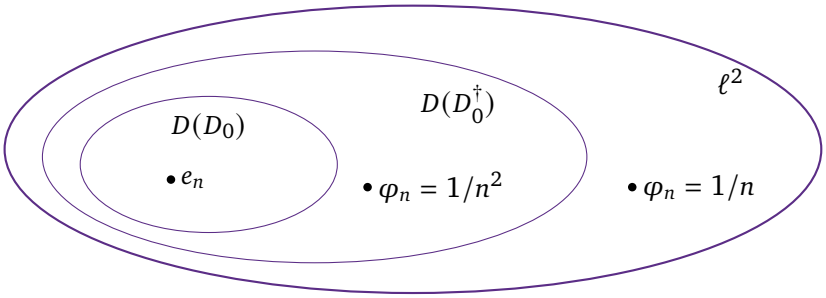


Figure 0.2: The domains of Example 0.6, drawn as nested sets, $D(D_0) \subsetneq D(D_0^\dagger) \subsetneq \ell^2$. Each dot is a sequence, drawn in the smallest of the three sets that contains it. The drawing shows inclusion, not distances in ℓ^2.

Self-check 0.2

1. Show that $\langle \psi, A\psi \rangle$ is real for every $\psi \in D(A)$ when A is symmetric.
2. Show that the sequence $\varphi_n = 1/n$ lies in ℓ^2 and not in $D(D_0^\dagger)$, as Figure 0.2 draws it.

0.3 How this course proves

Every result of this course carries one of three tags. A result with no tag is *proved* in full. A result tagged *sketched* has its central idea proved and routine details omitted, or it is proved with the help of a cited theorem. A result tagged *cited* is taken from a named source, with its page.

A few facts of analysis are used throughout. This course takes them without proof, as *standing facts*:

S1 *Plancherel*: the Fourier transform is unitary on $L^2(\mathbb{R}^n)$, with its inversion formula (Lecture 4).

S2 The Fourier basis $e^{2\pi i k x/L}/\sqrt{L}$ ($k \in \mathbb{Z}$) and the sine basis $\sqrt{2/L}\,\sin(n\pi x/L)$ ($n \geq 1$) of $L^2(0, L)$ are complete (Lecture 1 and Lecture 3).

S3 Products of orthonormal bases are orthonormal bases of $L^2(\mathbb{R}^2)$ and $L^2(\mathbb{R}^3)$ (Lecture 13).

S4 The smooth functions that vanish outside a closed bounded set inside the interval, or inside $\mathbb{R}^n$, are dense in L^2 of the interval, or of $\mathbb{R}^n$. So are the rapidly decreasing functions of Lecture 4.

Linear Algebra states S2, Plancherel on the line, and the density of the rapidly decreasing functions on the line. The other cases and S3 are adopted here. A result whose only unproved ingredients are standing facts counts as proved.

Self-check 0.3

1. A proof uses the sine basis theorem and a theorem cited from Teschl. Which tag does the result carry?

0.4 Multiplication operators

The diagonal operator of Example 0.6 multiplies the nth entry of a sequence by n. Its counterpart on a line multiplies a function by a function, and the position of a particle on a line multiplies by x (Lecture 4). The space $L^2(\mathbb{R})$ is defined in the same way as L^2 of an interval (*Linear Algebra*, Def. 13.15): its vectors are classes of square-integrable functions on $\mathbb{R}$, and two functions lie in the same class when they agree outside a set of measure zero (for example, a finite or countable set). For a bounded real multiplier on an interval, *Linear Algebra*, Ex. 14.6, shows that multiplication is a bounded self-adjoint operator. An unbounded multiplier, such as x on $\mathbb{R}$, needs a domain.

Multiplication by x shows why. The Gaussian $\psi(x) = e^{-x^2}$ lies in $L^2(\mathbb{R})$, and so does $x\psi(x)$, since $\int_{\mathbb{R}} x^2 e^{-2x^2}\,dx = \sqrt{2\pi}/8$ is finite. The function $\varphi(x) = (1+x^2)^{-1/2}$ also lies in $L^2(\mathbb{R})$, since $\int_{\mathbb{R}} dx/(1+x^2) = \pi$. But $|x\varphi(x)|^2 = x^2/(1+x^2)$ tends to 1 as $|x| \to \infty$, so its integral diverges. So $x\varphi$ does not lie in $L^2(\mathbb{R})$, and multiplication by x cannot act on the vector φ. Figure 0.3 draws the two pairs of functions. The next lemma takes as domain exactly the vectors that stay square-integrable.

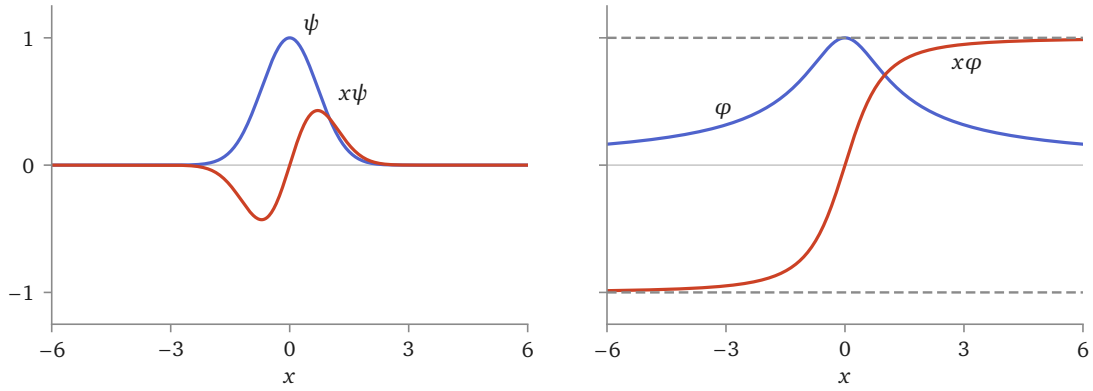


Figure 0.3: Left: the Gaussian $\psi(x) = e^{-x^2}$ and $x\psi(x)$, which both tend to 0. Right: $\varphi(x) = (1+x^2)^{-1/2}$ and $x\varphi(x)$, which tends to 1 as $x \to \infty$ and to -1 as $x \to -\infty$ (dashed lines). Whether a square-integrable function f lies in the domain of multiplication by x is decided by whether $\int_{\mathbb{R}} |xf(x)|^2 \, \mathrm{d}x$ is finite, not by the look of its graph.

Lemma 0.7 (Multiplication operators). *Let g be a real continuous function on $\mathbb{R}$, and let M_g act by $(M_g\psi)(x) = g(x)\psi(x)$ on $D(M_g) = \{\psi : g\psi \in L^2(\mathbb{R})\}$. Then M_g is self-adjoint. If g takes a value λ at only finitely many points, then λ is not an eigenvalue of M_g.*

Proof. Write 1_n for the indicator of $[-n, n]$. On $[-n, n]$ the continuous g is bounded.

1. *Densely defined.* For $\psi \in L^2(\mathbb{R})$, the vector $1_n\psi$ lies in $D(M_g)$. By the standing fact S4, for every $\epsilon > 0$ there is a smooth f that vanishes outside some $[-n_0, n_0]$ with $\|\psi - f\| < \epsilon$. For $n \geq n_0$, $\|\psi - 1_n\psi\| = \|(1 - 1_n)(\psi - f)\| \leq \|\psi - f\| < \epsilon$. So $1_n\psi \to \psi$.
2. *Symmetric*, since g is real.
3. *The adjoint.* Let $\varphi \in D(M_g^\dagger)$ and $\chi = M_g^\dagger\varphi$. The vector $\psi = 1_n(g\varphi - \chi)$ lies in $D(M_g)$, so by Definition 0.2
$$0 = \langle \varphi, M_g\psi \rangle - \langle \chi, \psi \rangle = \int_{-n}^{n} |g\varphi - \chi|^2 \, \mathrm{d}x.$$
So $1_n g\varphi = 1_n\chi$ for every n: the functions $g\varphi$ and χ agree on every $[-n, n]$, hence on $\mathbb{R}$, and $g\varphi$ is the class χ, which lies in $L^2(\mathbb{R})$. So $\varphi \in D(M_g)$ and $M_g^\dagger\varphi = M_g\varphi$. With the symmetry, $M_g^\dagger = M_g$.
4. *No eigenvalue.* Let $g\psi = \lambda\psi$. Let J be a closed bounded interval that contains none of the points where $g = \lambda$. On J the continuous $|g - \lambda|$ has a positive minimum c, so $c^2 \int_J |\psi|^2 \leq \int_J |(g - \lambda)\psi|^2 = 0$. The open intervals between consecutive points where $g = \lambda$, and the two outside them (or $\mathbb{R}$ itself, if there are no such points), are unions of such J, so ψ vanishes on each. They cover $\mathbb{R}$ except finitely many points, which do not change a class, so $\psi = 0$. ∎

Unlike D_0, which acts only on the finite sequences, M_g acts on every ψ for which $g\psi$ is square-integrable. A unitary change of variables keeps both conclusions of Lemma 0.7.

Lemma 0.8 (Unitary equivalence). *Let U be unitary and A self-adjoint on $\mathcal{H}$. The operator $B = U^{-1}AU$, on $D(B) = \{\psi : U\psi \in D(A)\}$, is self-adjoint, and it has the same eigenvalues as A.*

Proof. For $\psi \in \mathcal{H}$, take $a_j \in D(A)$ with $a_j \to U\psi$. Then $U^{-1}a_j \in D(B)$ and $U^{-1}a_j \to \psi$, so B is densely defined. Since $\langle U\varphi, U\psi\rangle = \langle \varphi, \psi\rangle$, and $\eta = U\psi$ runs through $D(A)$ as ψ runs through $D(B)$, the condition $\langle \varphi, B\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(B)$ reads $\langle U\varphi, A\eta\rangle = \langle U\chi, \eta\rangle$ for every $\eta \in D(A)$. So $\varphi \in D(B^\dagger)$ exactly when $U\varphi \in D(A^\dagger) = D(A)$, with $UB^\dagger\varphi = AU\varphi$. Hence $B^\dagger = B$. Finally, $B\psi = \lambda\psi$ if and only if $AU\psi = \lambda U\psi$. ∎

Lecture 4 applies the two lemmas with $g(x) = x$ for the position, and with the Fourier transform as U for the momentum and the energy.

Self-check 0.4

1. Let g be the constant function c. What is M_g, and which numbers are its eigenvalues? Which hypothesis of Lemma 0.7 fails?

Summary.

- *Domains are needed*: a symmetric operator defined on every vector is bounded. The position and the momentum of a particle cannot both be bounded, and the energy of a particle in a box is not (§ 0.1).
- *An operator* comes with its domain, and two operators are equal only when their domains are (Definition 0.1).
- *The adjoint* of a densely defined operator is defined on the vectors φ for which some χ satisfies $\langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(A)$ (Definition 0.2).
- *A symmetric operator* A satisfies $A \subseteq A^\dagger$. It is *self-adjoint* when the two are equal (Definition 0.3), and adjoints reverse inclusions (Proposition 0.5). The diagonal operator on finite sequences is symmetric, and its adjoint is larger (Example 0.6).
- *Multiplication* by a real continuous function, on the vectors it keeps square-integrable, is self-adjoint, and a value taken at only finitely many points is not an eigenvalue. A unitary equivalence keeps both (Lemmas 0.7 and 0.8).

Lecture 1 uses these definitions for the energy of a particle in a box, and Lecture 4 uses the multiplication operators for a particle on a line.

Exercises. Exercise Sheet 0.

Self-check answers.

0.2 1. Symmetry with $\varphi = \psi$ gives $\langle \psi, A\psi\rangle = \langle A\psi, \psi\rangle = \overline{\langle \psi, A\psi\rangle}$. 2. It lies in ℓ^2, since $\sum_n 1/n^2 < \infty$. It does not lie in $D(D_0^\dagger)$, since $\sum_n n^2/n^2$ diverges.

0.3 1. *Sketched*: the sine basis is a standing fact, but the theorem from Teschl is not.

0.4 1. $M_g\psi = c\psi$ for every $\psi \in L^2(\mathbb{R})$, so $M_g = c\mathbb{1}$. Every nonzero vector is an eigenvector with the eigenvalue c, and by Lemma 0.7 no other number is an eigenvalue, since g takes no other value. The function takes the value c at every point, not at finitely many.

References

Moretti, *Spectral Theory and Quantum Mechanics*, Prop. 11.32 (p. 627). Teschl, *Mathematical Methods in Quantum Mechanics*, Thm 2.9 (p. 67). *Linear Algebra for Physics*, Def. 13.15, Defs 14.2 and 14.3, Ex. 14.6. *Quantum Mechanics I*, Prop. 9.8.